

Coffee Shop Sales Analysis & Dashboard

1. Introduction

The **Coffee Shop Sales Dashboard** is an interactive data analytics project developed using Microsoft Excel. The purpose of the project is to analyze coffee shop sales data and transform raw transactional data into meaningful business insights through key performance indicators, charts, PivotTables, and interactive slicers.

The dashboard provides a consolidated view of sales performance across different stores, products, months, weekdays, and time periods.

2. Project Objective

The primary objective of this project is to understand the sales performance of the coffee shop and identify patterns that can support business decision-making.

The analysis focuses on answering questions such as:

- How much revenue is being generated?
- How many orders and items are being sold?
- Which store performs best?
- Which product categories generate the most sales?
- Which products are the top performers?
- Which days have higher sales?
- What time of the day generates the most sales?
- How do sales change across the months?

3. Tools & Technologies

The project was developed entirely using **Microsoft Excel**.

The following Excel features were used:

- Excel Tables
- Data cleaning and preparation
- Calculated columns
- Excel formulas
- PivotTables
- PivotCharts
- Slicers
- Data formatting
- Dashboard design
- Data visualization

A calculated **Time Period** column was also created to group transactions into broader periods such as Morning, Noon, Evening, and Night.

4. Key Performance Indicators

The dashboard contains four primary KPIs.

KPI	Value
Total Sales	\$698.81K
Total Orders	149,116
Total Items	214,470
Average Transaction Value	\$4.69

These KPIs provide a quick overview of the overall business performance before moving into detailed analysis.

5. Dashboard Analysis

5.1 Monthly Sales Trend

A line chart was used to analyze sales across the months.

The dashboard shows an overall upward movement in sales toward the later months of the dataset, with the highest monthly sales occurring toward the end of the displayed period.

This indicates that sales performance was not constant throughout the period and that monthly demand varied over time.

5.2 Sales by Store

Sales were compared across:

- Astoria
- Hell's Kitchen
- Lower Manhattan

Among the three locations, **Hell's Kitchen** records the highest sales, while the other two locations show slightly lower sales levels.

This comparison can help management investigate differences in customer demand and operational performance between locations.

5.3 Sales by Product Category

The category analysis shows that **Coffee is the strongest-performing product category** by sales.

Tea and Bakery also contribute significantly, while several smaller categories contribute comparatively less revenue.

This suggests that coffee products are a major contributor to overall sales and should receive appropriate attention in inventory planning and product management.

5.4 Sales by Weekday

Sales were analyzed across all seven days of the week.

The results show variation in sales throughout the week. Some weekdays generate stronger sales than others, while Saturday shows relatively lower sales compared with several weekdays in the displayed dataset.

This information can be useful for planning staffing levels and promotional activities.

5.5 Top 10 Products

The Top 10 Products chart identifies the individual products generating the highest sales.

Barista Espresso is the highest-performing product in the displayed analysis, followed by other products such as Brewed Chai Tea, Hot Chocolate, and Gourmet Brewed Coffee.

The analysis can help identify products that should receive priority in inventory planning and promotional strategies.

5.6 Sales by Time Period

Sales were grouped into:

- Morning
- Noon
- Evening
- Night

The analysis shows that **Morning is the strongest sales period**, followed by Noon and Evening, while Night records the lowest sales.

This indicates that customer demand is concentrated during earlier parts of the day.

6. Key Insights

The main findings from the dashboard are:

1. **Coffee is the largest product category by sales.**
2. **Hell's Kitchen has the highest sales among the three store locations.**
3. **Morning generates the highest sales among the four time periods.**
4. **Barista Espresso is the highest-performing product among the Top 10 products.**
5. **Monthly sales show an upward trend toward the later months of the dataset.**
6. **Sales vary across weekdays, indicating differences in customer demand throughout the week.**

7. Recommendations

Based on the analysis, the following recommendations can be considered:

1. Prioritize high-performing products

Ensure adequate inventory for high-demand products, particularly coffee products and the leading individual products.

2. Optimize staffing during peak hours

Since morning is the strongest sales period, staffing and operational resources can be aligned with higher customer demand during this period.

3. Analyze store-level differences

Hell's Kitchen demonstrates the highest sales. Management could investigate factors such as customer traffic, product demand, and operating practices that may contribute to the difference.

4. Use targeted promotions

Promotional campaigns can be considered for lower-performing time periods and weekdays to encourage additional customer visits and purchases.

5. Monitor monthly trends

Continued monitoring of monthly sales can help identify recurring patterns and support better inventory, staffing, and promotional planning.

8. Conclusion

The Coffee Shop Sales Dashboard demonstrates how Microsoft Excel can be used to convert transactional sales data into an interactive business intelligence tool.

By combining **KPIs, PivotTables, PivotCharts, calculated columns, and slicers**, the dashboard provides a simple way to explore sales performance from multiple perspectives.

The analysis highlights the importance of understanding **product demand, store performance, time-of-day patterns, weekday behavior, and monthly trends** when making operational decisions.

Overall, the project demonstrates practical application of Excel-based data analysis and visualization to a real-world business scenario.

9. Project Outcome

The completed dashboard provides:

Raw Data → Data Preparation → Analysis → Visualization → Insights → Recommendations

This workflow demonstrates the complete process of developing a basic business analytics dashboard using Excel.
